

Excelsior Ultrasound Institute LMS

Greenfield Canvas/WGU-hybrid learning platform for Excelsior Ultrasound Institute. Marketing site stays at excelsiorultrasoundinstitute.com; LMS on a subdomain (e.g. learn.excelsiorultrasoundinstitute.com).

Product decisions (locked)

- Scope v1: Full 18-month diploma structure first; ARDMS prep courses use the same engine (shorter programs).
- Model: Hybrid - modules -> lessons -> assignments/quizzes/exams + progress + gradebook (Canvas UX, WGU-like clarity of "where am I?").
- Roles: Owner, Dean, Instructor, Student.
- Marketing site: Keep excelsiorultrasoundinstitute.com as-is; LMS on a subdomain.
- Payments: Out of v1. Owner/Dean enroll students manually (fits ~20-student cohorts). Stripe can come later.

Recommended stack

- App: Next.js (App Router) on Vercel - Fast LMS UI, server actions, deploy-ready
- Auth: Clerk - Roles/orgs, invite flows, MFA for staff
- DB: Neon Postgres + Drizzle - Branchable DB, typed schema, migrations
- Files/video: Vercel Blob (+ optional Mux later) - PDFs, images; streaming can upgrade later
- UI: Tailwind + shadcn/ui - Clean Canvas-like density
- Email: Resend - Invites, grade notifications

Role boundaries (RBAC)

Enforced in DB (role on memberships) + server-side checks on every mutation (never UI-only).

- Owner - Can: everything (users, roles, programs, settings, audit, delete/archive). Cannot: n/a.
- Dean - Can: cohorts, enroll/unenroll, assign instructors, view all grades, unlock exams. Cannot: billing/system secrets, destroy Owner, global theme/domain.
- Instructor - Can: edit assigned course content, publish modules, grade, release quiz feedback. Cannot: enroll students, edit others courses, change roles.
- Student - Can: view enrolled content, submit work, quizzes/exams, own grades/progress. Cannot: see others grades, edit content.

Domain model (core)

Key entities:

- Program - e.g. Diagnostic Medical Sonography Diploma, ARDMS Abdominal Prep
- Cohort - dated intake (18-month class of $N \leq 20$)
- Course - Abdominal, OB/GYN, Vascular, Physics, etc.
- Module / Lesson - ordered content (text, PDF, video URL, embeds)
- Assessment - quiz | exam | assignment; timed optional; attempts; shuffle
- GradeEntry - weighted categories (quizzes, exams, clinical, participation)
- Progress - lesson completion + % through course

Seed content from public program: abdominal, OB/GYN, vascular, acoustic physics/instrumentation, cross-sectional anatomy, plus ethics/terminology/patient care.

App surfaces

- Student /app - dashboard (continue learning), course outline, lesson player, quiz/exam runner, grades, announcements
- Instructor /instructor - course builder, gradebook for assigned sections, quiz review
- Dean /dean - cohorts, enrollments, instructor assignment, program-wide grade overview, unlocks
- Owner /owner - all of the above + users/roles, program CRUD, platform settings, audit log

Shared shell: left nav by role, top current course/cohort context, dense but calm medical-education UI (brand from existing site).

Phased delivery

Phase 0 - Scaffold

- Init Next.js + Tailwind + shadcn, Clerk, Neon, Drizzle
- Auth middleware; role claim sync
- Empty dashboards per role; seed Owner account
- Deploy preview on Vercel; point subdomain when ready

Phase 1 - Academic skeleton

- Programs, courses, modules, lessons CRUD (Owner/Dean create; Instructor edit assigned)
- Enrollments + cohort management (Dean/Owner)
- Student read-only course outline + lesson viewer + mark-complete
- Progress bar per course

Phase 2 - Assessments + grades

- Quiz/exam builder (MCQ, multi-select, true/false; short answer manual grade)
- Timed exams, attempt limits, auto-score objective items
- Gradebook with category weights; student grade view
- Instructor grade override + feedback

Phase 3 - Owner depth + Dean ops

- User invite/disable; role assignment
- Course copy / template for new cohorts
- Announcements, basic audit log
- ARDMS prep as short Programs using same primitives
- Export grades CSV

Phase 4 - Later (not v1)

- Stripe (tuition deposit / prep course fees)
- Clinical hour logs / competency checklists
- Live proctoring, discussion forums, SCORM
- Video hosting (Mux), mobile apps

UX principles

- Canvas-like organization: course map, collapsing modules, next/prev lesson, breadcrumbs
- WGU-like fluidity: continue where you left off, % complete, what is due next
- Owner simplicity: wizards for New cohort -> copy courses -> assign instructors -> enroll students
- No role overstep: every action authorized server-side; UI hides what you cannot do

Repo starting point

Empty private repo Innoren/ExcelsiorUltrasoundInstitute. First commit will be the scaffolded app (no marketing site rewrite).

Success criteria for first usable release

- Owner can create diploma program + courses + invite Dean/Instructors/Students
- Dean can open a cohort and enroll ≤ 20 students
- Instructor can publish modules and a quiz; grade short answers
- Student can complete lessons, take a quiz, see grade and progress
- Roles cannot perform out-of-scope actions (verified by permission tests)

Implementation to-dos

- 1. Scaffold Next.js + Clerk + Neon/Drizzle + shadcn; role middleware; Vercel preview
- 2. Implement core schema (Program/Cohort/Course/Module/Lesson/Enrollment) + RBAC helpers
- 3. Build Owner/Dean consoles: users, programs, cohorts, enrollments, instructor assignment
- 4. Build Instructor course builder (modules/lessons) for assigned courses
- 5. Build Student dashboard, course outline, lesson player, progress
- 6. Quiz/exam engine, gradebook weights, student grade view, instructor overrides
- 7. Seed diploma program structure from Excelsior specialties; support short ARDMS prep programs
- 8. Permission tests, audit basics, CSV grade export, subdomain cutover checklist